Diffractive Beam Shaper Design via Finite Element Method

Mathematical Description of the Method

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1 The Problem

Design a **diffractive phase element** (DPE) with transmission function

$$T_{\mathrm{DPE}}(x, y) = \exp\bigl(i\,\phi(x, y)\bigr)$$

that transforms an input beam f_1 into a desired output intensity pattern $|f_2|^2$. Since $|T_{\mathrm{DPE}}| = 1$, no energy is absorbed. Wave propagation between the input and output planes (at focal distance f from a lens) is described by the 2D Fourier transform:

$$f_2 = \mathcal{F}\bigl\{T_{\mathrm{DPE}} \cdot f_1\bigr\}.$$

Since only the intensity $|f_2|^2$ matters (not its phase), there is a large design freedom that the algorithms below exploit.

2 The Geometric Transformation Approach

The central idea is to find a **geometric transformation** T that maps each point (x, y) in the input plane E to a point $T(x, y) = \bigl(T_x(x, y),\, T_y(x, y)\bigr)$ in the output plane A, such that T redistributes the intensity $|f_1|$ into $|f_2|$.

Once such a transformation is found, the **method of stationary phase** determines the phase function:

$$\nabla\phi(x, y) = \frac{2\pi}{\lambda f}\, T(x, y),$$

where λ is the wavelength and f is the focal length.

Since the phase of f_1 can be encoded in the DPE, we may assume without loss of generality that f_1 has constant phase. For f_2, only the intensity is prescribed (phase freedom).

2.1 Analytical Solutions

An analytical solution for the geometric transformation T exists in two cases:

1. **Separable to separable**: If $f_1(x, y) = g_1(x)\, h_1(y)$ and $f_2(x, y) = g_2(x)\, h_2(y)$, then T decomposes into two independent 1D problems. Each 1D transformation is found by **CDF matching**: T_x satisfies $\int_0^x |g_1(\xi)|^2\, d\xi = \int_0^{T_x(x)} |g_2(\xi)|^2\, d\xi$, and similarly for T_y.

2. **Isotropic to isotropic**: If both f_1 and f_2 depend only on the radial distance $r = \sqrt{x^2 + y^2}$, the problem reduces to a single 1D radial CDF-matching problem.

For signals that do not fall into these classes, no analytical solution is available.

2.2 The SepOp Baseline

The **Separabilisierungsoperator** (SepOp) maps any signal to a nearby separable signal:

$$\text{SepOp}\colon f(x,y) \;\mapsto\; \underbrace{\left(\int f(\xi,y)\,d\xi\right)}_{p(y)} \cdot \underbrace{\left(\int f(x,\xi)\,d\xi\right)}_{q(x)}.$$

In the discrete case: $q(x) = \sum_y f(x,y)$ and $p(y) = \sum_x f(x,y)$, giving $s(x,y) = q(x) \cdot p(y)$. After applying SepOp to both f_1 and f_2, the separable analytical method can be used. This provides a fast, parameter-free IFTA starting point, but can be inaccurate for highly non-separable signals.

2.3 The Finite Element Method

The FEM approach works for *arbitrary* input and output signals, including non-separable and non-isotropic ones. The idea:

1. Lay **topologically equivalent meshes** over planes E and A.
2. **Optimize node positions** so corresponding cells contain equal energy.
3. The node correspondences define a **discrete transformation** T.
4. **Recover** ϕ from T via polynomial least squares on $\nabla\phi$.

3 Rectangular Topology — Constant Intensity (Section 4.1)

Consider transforming a **circle with constant intensity** into a **square with constant intensity**. Since the intensity is constant, equal area implies equal energy.

3.1 Mesh Generation

Nodes are indexed by (i,j) with $i,j \in \{0,\dots,s-1\}$, forming a rectangular grid topology. The algorithm has three steps:

Step 1 — Boundary nodes. For the output (square), nodes are placed uniformly on the square boundary. For the input (circle), nodes are placed uniformly (by arc length) on the circle.

Step 2 — 4-point interpolation. Each interior node $P_0 = (i_0, j_0)$ is interpolated from its 4 associated boundary nodes $P_1, \dots, P_4$ (the boundary intersections along the i- and j-directions):

$$P_0 = \sum_{l=1}^{4} g_l\, P_l.$$

The weights depend on the Euclidean distances $d_l = [(i_0 - i_l)^2 + (j_0 - j_l)^2]^{1/2}$ in index space:

$$g_1 = \frac{d_2\, d_3\, d_4}{(d_1 + d_2)\,(d_1 d_2 + d_3 d_4)}, \qquad g_2 = \frac{d_1\, d_3\, d_4}{(d_1 + d_2)\,(d_1 d_2 + d_3 d_4)},$$
$$g_3 = \frac{d_1\, d_2\, d_4}{(d_3 + d_4)\,(d_3 d_4 + d_1 d_2)}, \qquad g_4 = \frac{d_1\, d_2\, d_3}{(d_3 + d_4)\,(d_3 d_4 + d_1 d_2)}.$$

Step 3 — Laplace smoothing. Interior nodes are smoothed using the discrete Laplace operator: each node moves toward the centroid of its 4 direct neighbors:

$$P_0' = P_0 + \alpha\,(C - P_0), \qquad C = \tfrac{1}{4}(P_1 + P_2 + P_3 + P_4).$$

3.2 Equal-Area Optimization (8-Point Algorithm)

For an interior node P_0 surrounded by 8 neighbors $P_1, \ldots, P_8$ (numbered counterclockwise: P_1=top, P_2=top-right, …, P_8=top-left), the **ideal position** $C(x_c, y_c)$ equalizing the 4 surrounding cell areas satisfies:

$$x_c = \frac{bf - de}{bc - ad}, \qquad y_c = \frac{af - ce}{bc - ad},$$

where

$$\begin{aligned}
a &= y_2 - y_8 - y_6 + y_4, & b &= x_2 - x_8 - x_6 + x_4,\\
c &= y_8 - y_6 - y_4 + y_2, & d &= x_8 - x_6 - x_4 + x_2,\\
e &= x_1(y_2-y_8) + x_5(y_4-y_6) + y_1(x_8-x_2) + y_5(x_6-x_4), & &\\
f &= x_3(y_2-y_4) + x_7(y_8-y_6) + y_3(x_4-x_2) + y_7(x_6-x_8). & &
\end{aligned}$$

The node is moved iteratively: $P_0' = P_0 + \alpha\,(C - P_0)$. A validity check rejects moves that would create negative-area (tangled) cells.

Boundary nodes on the circle are moved along the circle arc to equalize their two adjacent cells, using the 5-point boundary algorithm.

4 Triangular Topology — Constant Intensity (Section 4.2)

Consider transforming a **circle with constant intensity** into an **equilateral triangle with constant intensity**. The triangular topology is more natural for this geometry.

4.1 Mesh Generation

Nodes are indexed by (i, j) with $j = 0, \ldots, s{-}1$ and $i = 0, \ldots, j$. The total number of nodes is $s(s{+}1)/2$. Cells are triangles.

Interior nodes are initialized using the **6-point interpolation** formula:

$$P_0 = \sum_{l=1}^{6} g_l\, P_l,$$

where $P_1, \ldots, P_6$ are the 6 associated boundary nodes along the 3 triangle directions, and the weights have the same distance-based structure as the 4-point formula but with 3 pairs of directions.

4.2 Equal-Area Optimization (9-Point Algorithm)

The 8-point algorithm from Section 4.1 is not adapted to triangular topology. Instead, the **9-point algorithm** is used. An interior node P_0 has 9 surrounding neighbors $P_1, \ldots, P_9$, forming triangular cells $A, B, \ldots, I$. The equal-area conditions are:

$$D + H + I = B + E + F, \tag{1}$$

$$G + H + I = A + C + B. \tag{2}$$

These give two linear equations for the ideal position (x_c, y_c):

$$x_c = \frac{ed - fb}{ad - cb}, \qquad y_c = \frac{af - ce}{ad - cb},$$

with coefficients $a, \ldots, f$ that are linear combinations of the 9 neighbor coordinates (see the Studienarbeit for full formulas).

5 General Signals — Arbitrary Intensity (Section 4.3)

For signals with **non-uniform intensity**, we use rectangular topology but weight the optimization by energy content.

The mesh starts as a uniform square grid on $[0,1]^2$. In each iteration:

1. Compute the energy in each cell (by scanning pixels inside the polygon using a ray-casting point-in-polygon test).
2. Move each interior node toward the **energy-weighted centroid** of its 4 direct neighbors:
$$C = \frac{w_L\, P_L + w_R\, P_R + w_U\, P_U + w_D\, P_D}{w_L + w_R + w_U + w_D},$$
where w_L is the total energy in the two cells to the left of the node, etc.
3. A validity check rejects moves that would tangle the mesh.
4. Boundary nodes are moved along their edge to equalize adjacent cells.

This naturally clusters nodes in high-energy regions (small cells) and spreads them in low-energy regions (large cells).

6 Phase Recovery (Section 4.4)

Given optimized mesh correspondences $\text{in}(i,j) = (x_{ij}, y_{ij})$ and $\text{out}(i,j) = (\phi_x^{ij}, \phi_y^{ij})$, we recover the phase function.

The stationary phase condition is:

$$\nabla\phi(x_{ij}, y_{ij}) = c \cdot \begin{pmatrix} \phi_x^{ij} - \frac{1}{2} \\ \phi_y^{ij} - \frac{1}{2} \end{pmatrix}, \qquad c = 2\pi N,$$

where N is the FFT grid size. We approximate ϕ as a bivariate polynomial of degree $D{-}1$:

$$\phi(x,y) = \sum_{\substack{k,l \geq 0 \\ k+l \leq D-1}} a_{kl}\, x^k y^l,$$

with $D(D{+}1)/2$ unknown coefficients. Setting $\partial g / \partial a_{kl} = 0$ for the least-squares objective gives a linear system $M\mathbf{a} = \mathbf{b}$.

7 Quality Metrics (Section 5)

Signal-to-Noise Ratio:

$$\text{SNR}_I(f_2, f) = \frac{\|f\|^2}{\||f_2| - \alpha|f|\|^2}, \qquad \alpha = \frac{\langle |f|, |f_2| \rangle}{\|f\|^2}.$$

Diffraction Efficiency:

$$\eta = \frac{\|f\|^2_{W_{\text{Signal}}}}{\|f_1\|^2_{W_{\text{DE}}}}.$$

8 Original Results (1997)

Input → Output	**Method**	**SNR (dB)**	**η (%)**
Circle → Marilyn	SepOp	19.3	78.6
	Random	13.8	68.9
	FEM	**20.8**	**79.0**
Insep. → Marilyn	SepOp	32.8	92.1
	FEM	**35.8**	86.0
Gauss → Marilyn	**SepOp**	**31.4**	**92.5**
	FEM	29.9	83.0

9 References

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